Total time for the task: 50 minutes
Total marks: 50 marks
Weighting: 15% of the school mark

Question 1 **(23 marks)**

A study was undertaken into the factors affecting the reaction times of drivers. Three young drivers (Ella, Hiro and Joseph) who had recently obtained their drivers licences, participated in the study. Results recorded in the study included the amount of time it took for the participants to react to a hazard (reaction time) under various conditions that could cause driver impairment. The results of the study are shown below.

Driver reaction time under various conditions

Travelling at an average driving speed of 60 km/h	**Reaction time (s)**			
	Ella	Hiro	Joseph	Mean
Baseline (under normal conditions)	0.45	0.49	0.54	0.49
Reading a text message	0.79	0.83	1.12	
Writing a text message	0.91	0.86	1.04	
At 0.08 blood alcohol concentration	0.76	0.79	0.94	

The following questions refer to the data from the study outlined above.

(a) Formulate a hypothesis for the study. (2 marks)

(b) Calculate the mean reaction times for reading a text message, writing a text message and at 0.08 blood alcohol concentration. Insert the answer into the blank cells in the table above. (3 marks)

(c) The data on reaction time were recorded in seconds to two decimal places. Explain why this was used as the unit for measurement rather than minutes. (2 marks)

(d) Suggest **three** possible variables that would have been controlled for the study to be a fair test.
(3 marks)

One:

Two:

Three:

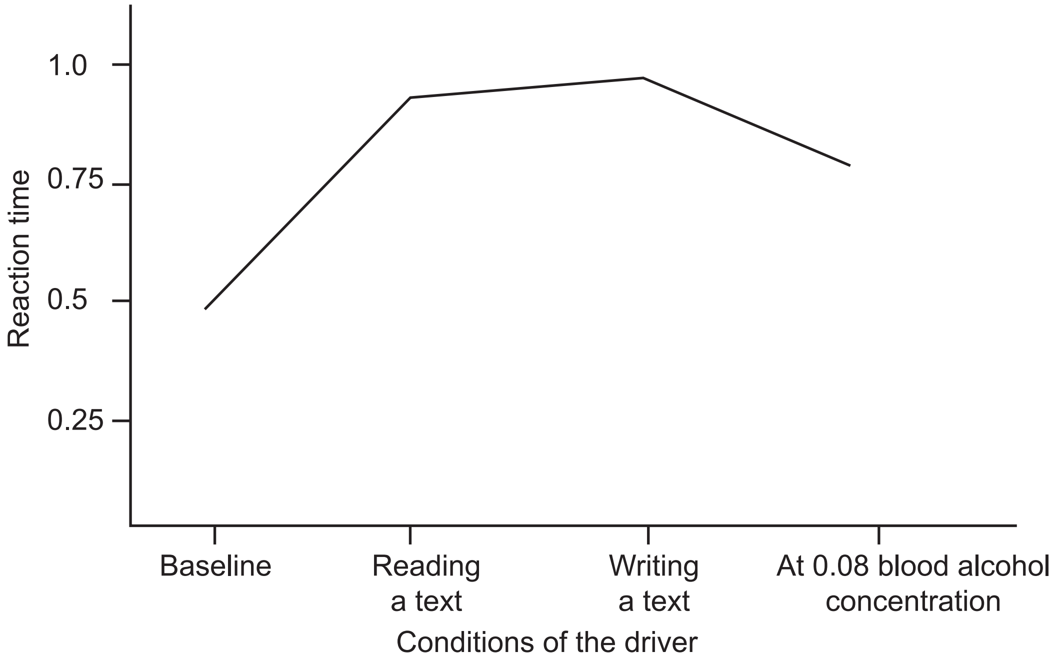
(e) Why was the baseline measurement of reaction time taken? (1 mark)

(f) Suggest **one** way in which the study could be modified to increase its

(i) reliability. (1 mark)

(ii) validity. (1 mark)

(g) One of the students attempted to graph the mean results. The graph is shown below.



There are three major errors in the graph above. Identify the **three** errors. (3 marks)

One: ____

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Two: ____

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Three: ____

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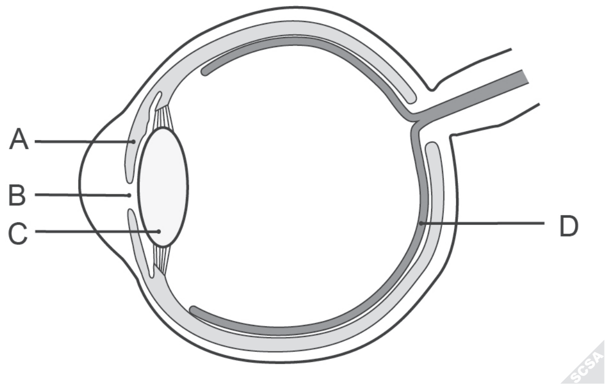
(h) Graph the mean data from the table correctly. (5 marks)

(i) With reference to the data obtained from the study, describe a reasonable conclusion that can be drawn. (2 marks)

Question 2 **(14 marks)**

Numerous hazards occur during every road journey. The body must detect, interpret and react to these hazards to ensure that no accidents occur. This requires the integration of many systems. The following questions refer to these systems.

One of the first steps in reacting to a hazard is its visual detection by the eye. Parts (a) and (b) of the question refer to the diagram of the eye shown below.



(a) For the structures listed below, outline the function of each as it relates to the detection of visual stimuli. (3 marks)

A: ____________________

B: ____________________

C: ____________________

(b) Describe how light hitting structure D is transmitted to the brain. (2 marks)

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